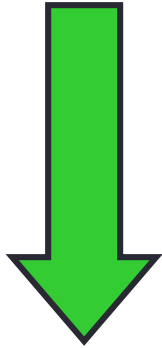


THE KCP LIMITED

Personal Protective Equipment

Hierarchy of Controls

Most
Effective



Least
Effective

Engineering

Administrative

Personal
Protective
Equipment

Personal Protective Equipment Program

- To ensure that employees have appropriate PPE for each job task with hazards to mitigate risk
- Complexity of program varies with types of hazards and size of the organization
- Programs typically include the following nine elements

1 — Hazard Assessment

- Job task analysis
 - Identify each step
 - Assess the magnitude of the hazard(s) at each step
 - Include other workers that may be affected
 - Determine if PPE is necessary as well as feasibility of engineering / administrative controls

Identify All Routes of Entry

- Consider all parts of the worker's body
 - Inhalation
 - Skin absorption
 - Ocular
 - Ingestion
 - Injection



(Source: NIOSH)

2 — PPE Selection

- PPE must
 - provide necessary protection
 - not restrict the worker from doing the task
 - Sufficient dexterity
 - not create new hazards
 - new hazards could include heat stress, ergonomic issues, communication interference, limited mobility, etc.
 - provide a degree of comfort (so that it will be used)

3 — Develop Standard Operating Procedures

- Typically written and includes:
 - specificity of what PPE is required for what hazard and when it is to be used
 - when replacement of PPE is necessary to prevent breakthrough or degradation
 - care of PPE

4 — Fitting PPE to the Worker

- Proper fit is critical for both effective protection and continued use.
- Adjustable size features should be identified and workers trained in the appropriate adjustments.

5 — Training

So workers know

- When to use PPE (which task or operation)
- The magnitude of the hazard and what PPE to use based on the hazard assessment
- How to inspect PPE for damage or non-functioning issues
- How to put it on and take it off (don and doff)
- When to replace the PPE
- How to check fit and effectiveness
- How to store and maintain when not in use

6 — Maintenance of PPE

- Maintenance – see manufacturer's recommendations
- Inspection – wear and damage
- Storage – to assure effectiveness when needed
- Cleaning – to keep clean and sanitary (or alternatively use disposable PPE)
- Shelf life – often specified by manufacturer

7 — Employer Provided and Employee-Owned PPE

- Normally employers are required to provide/pay for PPE although OSHA allows some exceptions (non-specialty safety-toe footwear, etc.).
- Employers are also responsible for ensuring that *employee-owned* PPE meets necessary protection and prohibit its use when it is defective or unsanitary.



(Source: OSHA)

8 — Recordkeeping

- Additional records should be maintained for:
 - PPE selection and worker training program
 - Written certification for the workplace PPE/hazard assessment (required by OSHA)

Voluntary Use of PPE

- Includes situations where workers choose to utilize PPE even though its use is not required by the hazard assessment.
- Employer maintains the responsibility that PPE is used correctly and does not pose additional hazards.

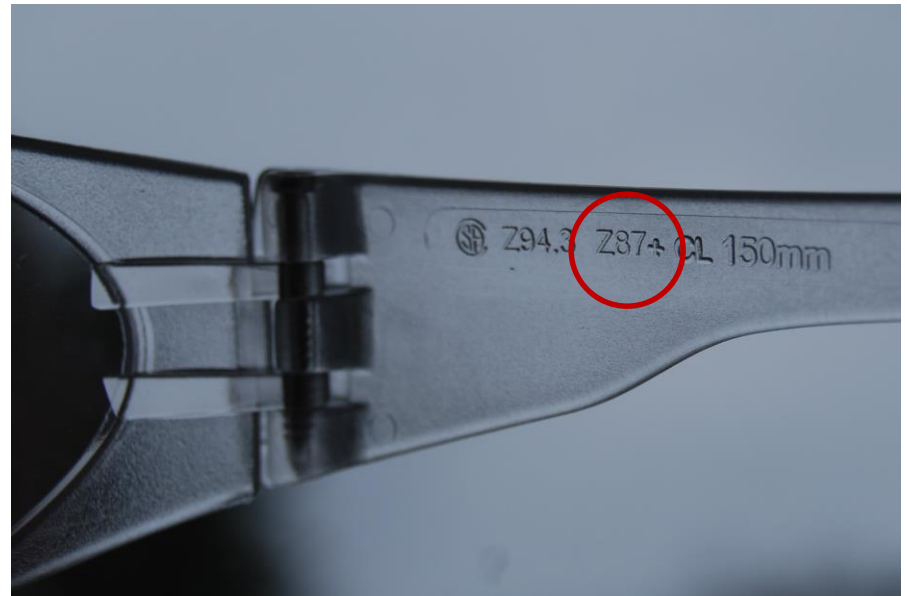
9 — Program Evaluation

- Annual evaluation based on:
 - experience and accidents
 - changes in work environment
 - changes in PPE
 - evaluation to assure PPE continues to provide necessary protection
 - complaints and feedback from users

Specific Personal Protective Equipment

Eye and Face Protection

- Impact-resistant eyewear for protection against flying objects (chips and particles)
 - marked with a plus sign (e.g., Z 87+ or Z 87-2+, if prescription)
 - integral lateral protection (side shields)



Reference: ANSI Z87.1

Eye and Face — Dust and Chemicals

- Designed to give additional protection from dust or droplets typically utilizing a seal against the face
- Chemical face shield worn over spectacles
- Chemical goggles; marked as
 - Splash and droplet protection (D3)
 - Dust (D4)
 - Fine dust (D5)
- Chemical face shield worn over goggles
- Full-face respirator



Courtesy MSA

Eye and Face — Optical Radiation Hazards

- Protect against infrared, ultraviolet and visible light
- Include direct or reflected light
- Marked “W” for welding, “R” for infrared, “U” for ultraviolet and “L” for visible light and glare
- Shade numbers indicate the optical density

(Source: NIOSH)



Head Protection — Type and Class

- ANSI Z89.1
 - Type I – Impact or penetration protection from above
 - Type II Additional protection from side, front, rear or off-center
 - Class C (conductive but no electrical protection)
 - Class G – tested to 2,200 V
 - Class E – tested to 20,000 V



Type 1, Class E
Courtesy MSA

Head Protection — Optional Features

- Reverse donning—must be designed for that feature—see manufacturer's literature
- High visibility choice
- Option for mounting for additional PPE: face shield, ear muffs, goggles, head-lamps, etc.



Head protection with ear muffs

Head Protection Use

- Fitting
- Replace if damaged, cracked, chalked from UV (sun) exposure
- Decals, painting—see manufacturer recommendations
- Service life of head protection—see manufacturer recommendations

Foot Protection — Electrical Hazards

- Electrical shock resistant shoes—tested to 14,000 V (dry conditions)
- Static dissipative—some electrical shock protection
- Conductive footwear—minimize the build-up of electrical charge
 - See ASTM F1116, F1117

Foot Protection — Design Options

- ASTM F-2412, F2413
 - Impact and compression resistant (toe caps) for falling or rolling objects.
 - Protection against objects piercing the sole
- Chain saw cutting resistance
 - ASTM F1458, F1818
- Metatarsal protection
- Slip resistance

Hand Protection

- Hazards that potentially can be reduced by gloves/PPE
 - abrasion
 - temperature extremes
 - puncture/cuts
 - biological (infectious)
 - electrical
 - radiation
 - vibration
 - low visibility (where high reflectivity is needed)
 - chemical

Hand Protection

- When choosing hand protection, consider:
 - Size/fit
 - Potential for glove material to be caught in machinery
 - Loss of dexterity and tactile feedback
 - Sensitization (e.g., latex) and dermatitis (from certain chemical accelerators used in rubber glove manufacture)
 - Internal contamination resulting in occlusion of chemical against the skin
 - Need to frequently assure protection integrity of the material (e.g., electrical gloves and sleeves)

Chemical-Protection Clothing/PPE

- Hazards of chemical contact potentially include:
 - significant absorption hazard leading to systemic effects
 - “skin” notation in ACGIH TLVs or OSHA PELs indicates dermal absorption warning
 - tissue destruction (local effects)
 - necrosis of tissue, e.g., corrosives, caustics
 - dermal sensitivity (allergic dermatitis)
 - includes chemicals with D-SEN notation in ACGIH TLVs



Cement dermatitis
(Source: NIOSH)

Chemical-Protection Clothing

- Decisions on type of protective clothing and the choice of material from which it is made depend on:
 - job task requirements
 - chemicals involved
 - integration of other PPE
 - potential to create additional hazards (ergonomic, heat stress)

Chemical-Protection Clothing (cont.)

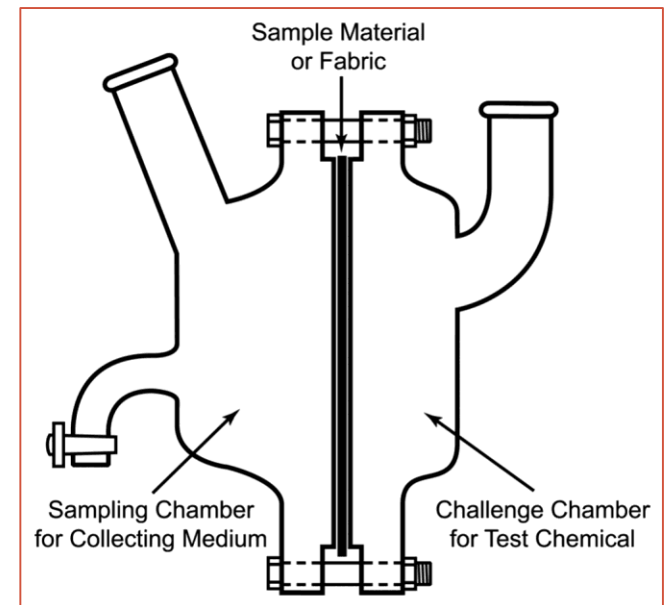
- Effectiveness:
 - No single material/fabric offers protection against all chemicals nor *unlimited* protection against any specific chemical
- Concerns:
 - *penetration* of chemicals through clothing or ensemble mechanics such as through seams, zippers, stitching, etc.
 - *degradation*—damage caused by chemical on the fabric
 - *permeation* of chemicals through fabric at molecular level

Chemical-Protection Clothing (cont.)

- Penetration of chemicals
 - fabric seams provide varying amounts of protection depending on type of connection (stitched, cemented, heat sealed, etc.)
- Tape
 - keeps seams or ensemble components in proper alignment
 - some chemical-resistant tapes have been tested
 - not appropriate for rip repairs on material/fabric

Chemical-Protection Clothing (cont.)

- Permeation: chemical specific
 - Manufacturers' permeation test typically measured by:
 - Break-through detection time (hours)
 - Permeation rate ($\mu\text{g}/\text{cm}^2/\text{min}$)
 - Cumulative break-through in 1 hour (gm or drops per hour)



Permeation test cell

Levels of Protection

Level	Protective suit
A	Totally encapsulating, vapor-protective suit
B	Hooded chemical resistant clothing
C	Hooded chemical resistant clothing
D	Coveralls

PPE and Ionizing Radiation

- PPE must be able to reduce the penetration and the workers' exposure to ionizing radiation by absorbing the energy in the PPE material.
 - choice dependent on the type and magnitude of the radiation
 - Not feasible for several types of radiation, e.g., high energy x-ray or gamma
 - protection dependent on the density and thickness of the PPE material